Derivative Securities FNCE 30007

Tutorial 7: BSM Model Part 1

Zhibo Zhang

Department of Finance
University of Melbourne

September 2026

Review

Notation:

- μ: the expected return on the stock.
- σ: the volatility of the stock, defined as the standard deviation of continuously compounded returns over one year.

If the short-period return is normally distributed, then $\log S_T$ is normally distributed; equivalently, S_T is lognormally distributed.

- $$\log S_T \sim \mathcal{N}\big(\log S_0 + (\mu - \tfrac{1}{2}\sigma^2)T,\, \sigma^2 T\big)\,.$$

Tutorial Question 1(a)

Q1: Let X be a normally distributed random variable with mean 100 and variance 100.

(**a**) What is the probability that X will take on a value less than 85?

Standardize X: $z = \frac{x-\mu}{\sigma}$, where $z \sim N(0,1)$. Since $\mu = 100$ and $\sigma = 10$,

$$z = \frac{85 - 100}{10} = -1.5.$$

Therefore,

$$\Pr(X < 85) = \Pr(z < -1.5) = 0.0668.$$

Tutorial Question 1(b)

Q1: Let X be a normally distributed random variable with mean 100 and variance 100.

(**b**) What is the probability that X will take on a value greater than 90?

If $X = 90$, then

$$z = \frac{90 - 100}{10} = -1.$$

Therefore,

$$\Pr(X > 90) = \Pr(z > -1) = 0.8413.$$

Tutorial Question 1(c)

Q1: Let X be a normally distributed random variable with mean 100 and variance 100.

(c) What is the probability that X will take on a value less than 85 or more than 90?

$$\begin{aligned}
\Pr(X < 85 \text{ or } X > 90) &= \Pr(X < 85) + \Pr(X > 90) \\
&= \Pr(z < -1.5) + \Pr(z > -1) \\
&= 0.0668 + 0.8413 \\
&= 0.9081.
\end{aligned}$$

Tutorial Question 2

Q2: Let Z be a standard normal random variable.

(a) What is the probability that Z will take on a value between -1 and 1?

(b) What is the probability that Z will take on a value between -2 and 2?

(c) Comment on your answers in parts (a) and (b).

Note that the normal distribution is symmetric:

(a) $\Pr(-1 < z < 1) = N(1) - N(-1) = 0.6826$.

(b) $\Pr(-2 < z < 2) = N(2) - N(-2) = 0.9544$.

(c) About 68% of observations are within one standard deviation, and about 95% are within two standard deviations.

Tutorial Question 3

Q3: The current price of a stock is \$50. Assume the expected return on the stock is 18% p.a. and its volatility is 30% p.a. In the Black–Scholes framework, what is the probability distribution of the stock price in:

(a) one year's time?

(b) two years' time?

$$\log S_T \sim \mathcal{N}\big(\log S_0 + (\mu - \tfrac{1}{2}\sigma^2)T,\, \sigma^2 T\big)\,.$$

(a) $\log S_T \sim \mathcal{N}\big(\log 50 + (0.18 - 0.5 \times 0.3^2),\, 0.3^2\big)$.

(b) $\log S_T \sim \mathcal{N}\big(\log 50 + (0.18 - 0.5 \times 0.3^2) \times 2,\, 0.3^2 \times 2\big)$.

Review: BSM Model

The Black–Scholes–Merton formula for a European option written on a non-dividend-paying stock is:

$$c = S_0 N(d_1) - K e^{-rT} N(d_2),$$
$$p = K e^{-rT} N(-d_2) - S_0 N(-d_1),$$

where

$$d_1 = \frac{\log(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}},$$
$$d_2 = d_1 - \sigma\sqrt{T}.$$

Review: Intuition for the Call Formula

Without going through the full proof, we can check whether the formula behaves sensibly.

- If $S_0 \to 0$, then $\log(S_0/K) \to -\infty$, so $d_1, d_2 \to -\infty$. Then $N(d_1), N(d_2) \to 0$ and $c \to 0$.
- If S_0 is very high, then d_1, d_2 are very positive. Then $N(d_1), N(d_2) \to 1$ and $c \to S_0 - Ke^{-rT}$.
- If $T \to 0$ as well, this approaches the call payoff $\max(S_0 - K, 0)$.

Tutorial Question 4

Q4: Consider a non-dividend-paying stock with current price \$10.00, risk-free interest rate 6% p.a. and volatility 40% p.a.

(a) Use the Black–Scholes call option formula to calculate the current price of a 2-year European call option with strike \$10.00.

(b) Use the Black–Scholes put option formula to calculate the current price of a 2-year European put option with strike \$10.00.

(c) Use the simulator to check your answers.

Tutorial Question 4(a): Call

We summarize the information:

$$S_0 = 10, \qquad K = 10, \qquad r = 0.06, \qquad \sigma = 0.4, \qquad T = 2.$$

$$\begin{aligned} d_1 &= \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \\ &= \frac{0 + (0.06 + 0.5 \times 0.4^2) \times 2}{0.4\sqrt{2}} \approx 0.495, \\ d_2 &= d_1 - \sigma\sqrt{T} \approx 0.495 - 0.4\sqrt{2} \approx -0.071. \end{aligned}$$

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \approx 10N(0.495) - 10e^{-0.12} N(-0.071) \approx \boxed{2.71}.$$

Tutorial Question 4(b): Put

Using the same $d_1 \approx 0.495$ and $d_2 \approx -0.071$,

$$P = Ke^{-rT}N(-d_2) - S_0N(-d_1) \approx 10e^{-0.12}N(0.071) - 10N(-0.495) \approx \boxed{1.58}.$$

Check using put–call parity:

$$C + Ke^{-rT} = P + S_0,$$

so

$$P = C + Ke^{-rT} - S_0 \approx 2.71 + 10e^{-0.12} - 10 \approx 1.58.$$

Tutorial Question 5

Q5: Reconsider the European call option in Q4(a). Recalculate the price of the option using:

(a) a one-period binomial model;

(b) a ten-period binomial model using the simulator;

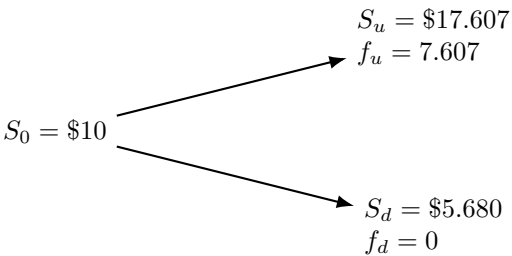
(c) what do your answers in parts (a), (b) and Q4(a) suggest about the relationship between the binomial and Black–Scholes models?

Tutorial Question 5(a): One-Period Binomial

For the one-period model, $\Delta t = T = 2$.

$$u = e^{\sigma\sqrt{\Delta t}} = 1.7607, \qquad d = e^{-\sigma\sqrt{\Delta t}} = 0.5680.$$

$$p = \frac{e^{r\Delta t} - d}{u - d} = 0.4691.$$



$$c = e^{-r\Delta t}\big(pf_u + (1-p)f_d\big) \approx \boxed{3.16}.$$

Tutorial Question 5(b): Ten-Period Binomial

For the ten-period model, $\Delta t = T/10 = 0.2$.

$$u = e^{0.4\sqrt{0.2}} = 1.1959, \qquad d = e^{-0.4\sqrt{0.2}} = 0.8362,$$

$$p = \frac{e^{0.06\times 0.2} - d}{u - d} = 0.4890.$$

For a European call, use terminal payoffs and discount the risk-neutral expectation:

$$c = e^{-rT} \sum_{j=0}^{10} \binom{10}{j} p^j (1-p)^{10-j} \max\left(S_0 u^j d^{10-j} - K, 0\right).$$

Using the simulator/backward induction:

$$\boxed{c_{10} \approx 2.66}.$$

Tutorial Question 5(c): Interpretation

Method	Number of periods	Call value
Binomial	1	3.16
Binomial	10	2.66
Black–Scholes	continuous limit	2.71

The binomial model approaches the Black–Scholes value as the number of periods increases. In finite trees the estimate can oscillate around the limit, especially when comparing even and odd numbers of steps.

Link Between Binomial and BSM

Both the binomial tree and BSM value an option as the present value of its expected payoff under the risk-neutral measure.

- The binomial model uses a discrete distribution with a finite number of stock-price outcomes.
- BSM is the continuous-time limit of this idea.
- As $n \to \infty$ and $\Delta t = T/n \to 0$, the binomial distribution of log returns converges to the normal distribution assumed in BSM.